

TRUESMILES DENTAL CLINIC MANAGEMENT SOFTWARE REQUIREMENTS DOCUMENT

Overview

Develop a modern, secure, cloud-based Dental Clinic Management Software FOR TRUESMILES DENTAL CLINIC operating three branches:

Woji Branch – Port Harcourt

Eneka Branch – Port Harcourt

Bonny Island Branch

The system must be accessible via Web and Mobile devices, with a clean, modern, user-friendly interface that simplifies patient management, appointment scheduling, billing, reporting, and branch performance monitoring.



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PROJECT OBJECTIVES

The software should:

Improve patient experience.

Reduce administrative workload.

Centralize operations across all branches.

Enable real-time monitoring by management.

Improve patient record management.

Increase operational efficiency.

Provide detailed business intelligence and analytics.

USER ROLES

1. MASTER ADMINISTRATOR

The Master Administrator should have access to all branches and all system functionalities.

Permissions

View all branches.

View all patients.



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View all appointments.

View all financial transactions.

View all doctors and nurses.

Generate reports.

Manage staff accounts.

Configure system settings.

Receive automated reports.

Monitor branch performance.

2. BRANCH ADMINISTRATOR

Each branch should have its own administrator.

Permissions

Manage branch operations.



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Register patients.

Manage appointments.

Manage staff.

Generate branch reports.

Manage billing.

Track patient attendance.

3. DOCTOR PORTAL

Features

View daily appointment schedule.

Access patient records.

Review nurse notes.

Record diagnosis.

Create treatment plans.



Prescribe medication.

Request investigations.

Upload X-rays and dental images.

Update treatment progress.

Approve treatment completion.

4. NURSE PORTAL

Features

View incoming appointments.

Register walk-in patients.

Record patient vital signs.

Take preliminary medical history.

Upload patient information.



Forward patient information to assigned doctor.

Monitor waiting patients.

Update consultation status.

Schedule follow-up appointments.

5. RECEPTIONIST PORTAL

Features

Register patients.

Schedule appointments.

Print invoices.

Verify payments.

Manage waiting queue.

Send reminders.



6. PATIENT PORTAL

Features

Create account.

Login securely.

View clinic branches.

View available doctors.

View available nurses.

Book appointments online.

Reschedule appointments.

Cancel appointments.

View treatment history.

View prescriptions.

Download invoices.



Make online payments.

Receive notifications.

Chat with clinic support.

PATIENT MANAGEMENT MODULE

Online Registration

Patients should be able to:

Register online.

Upload profile picture.

Upload identification documents.

Enter medical history.

Automatic Patient ID Generation

Every registered patient must automatically receive a unique Patient ID.

Example:



WJ-2026-00001

EN-2026-00001

BN-2026-00001

Patient ID must never duplicate.

APPOINTMENT MANAGEMENT

Features

Online booking.

Walk-in registration.

Appointment approval.

Doctor availability management.

Automated scheduling.

Queue management.



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Appointment reminders.

Notifications

Automatic reminders via:

SMS

Email

WhatsApp (optional)

Reminders should be sent:

24 hours before appointment.

3 hours before appointment.

Immediately after booking confirmation.

BRANCH INFORMATION DISPLAY

Patients should be able to see:



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Number of doctors available.

Number of nurses available.

Branch operating hours.

Branch location.

Contact information.

Current queue status.

Available consultation slots.

ELECTRONIC MEDICAL RECORD (EMR)

The system should maintain complete patient records.

Data Stored

Personal details.

Medical history.

Dental history.



Allergies.

Treatment records.

Prescriptions.

X-ray records.

Laboratory results.

Progress notes.

Follow-up records.

BILLING & PAYMENT MODULE

Features

Automatic invoice generation.

Consultation fees.

Treatment charges.



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Procedure charges.

Medication charges.

Online Payment Options

Debit Card

Bank Transfer

USSD

Mobile Money

Payment Gateway Integration

Payment Tracking

Paid

Partially Paid

Unpaid

Overdue



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Patients should receive:

Receipt by email.

Receipt by SMS.

Downloadable PDF invoice.

DAILY OPERATIONS REPORTING

The software should automatically record:

Total patients attended daily.

Total consultations.

Total treatments.

Total revenue.

Outstanding payments.



AUTOMATED WEEKLY REPORTS

Every week, the Master Administrator should receive email reports containing:

Branch Performance

Total patients attended.

Total appointments booked.

Appointment completion rate.

Revenue generated.

Outstanding debts.

Most active doctor.

Most active nurse.

Reports should be available in:

PDF

Excel



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ANALYTICS & BUSINESS INTELLIGENCE DASHBOARD

The system should automatically generate:

Branch Analysis

Most busy branch.

Branch with highest patient count.

Branch with highest revenue.

Branch growth trend.

Average daily attendance.

Patient retention rate.

Doctor Analysis

Number of patients seen.

Revenue generated.



Treatment success rates.

Patient Analysis

New patients.

Returning patients.

Missed appointments.

Treatment frequency.

Graphical Dashboard

Charts

Pie Charts

Bar Graphs

Trend Analysis

INVENTORY MANAGEMENT MODULE

Features



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Track:

Dental materials.

Equipment.

Consumables.

Medications.

Alerts

Automatic alerts when stock:

Is low.

Expires soon.

Is out of stock.

HUMAN RESOURCE MODULE

Staff Management



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Store:

Staff records.

Qualifications.

Certifications.

Attendance.

Leave requests.

Performance Tracking

Attendance reports.

Productivity reports.

Monthly performance ratings.

PATIENT COMMUNICATION SYSTEM

Automated Messages



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Appointment reminders.

Payment reminders.

Treatment follow-ups.

Birthday greetings.

Oral health tips.

DOCUMENT MANAGEMENT

Upload and store:

X-rays.

Scan results.

Referral letters.

Prescriptions.

Consent forms.



FEEDBACK & REVIEW MODULE

Patients should be able to:

Rate services.

Submit complaints.

Provide suggestions.

Management dashboard should display:

Satisfaction score.

Complaint trends.

Service quality rating.

SECURITY REQUIREMENTS

Security Features

Multi-factor authentication.



Role-based access control.

Data encryption.

Automatic backups.

Audit logs.

GDPR-compliant architecture.

HIPAA-inspired patient privacy standards.

MOBILE APPLICATION

The system should include:

Patient App

Appointment booking.

Payment.

Notifications.

Medical records access.



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Staff App

Schedule viewing.

Patient management.

Reporting.

AI & SMART FEATURES (HIGHLY RECOMMENDED)

AI Analytics

Predict busy periods.

Predict patient no-shows.

Revenue forecasting.

Staff workload balancing.

Smart Reminders

Automatically suggest:



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Follow-up appointments.

Teeth cleaning schedules.

Routine dental checkups.

SYSTEM PERFORMANCE REQUIREMENTS

Fast loading time (<3 seconds).

99.9% uptime.

Cloud-based architecture.

Scalable for future branches.

Responsive on desktop, tablet, and mobile.

FUTURE EXPANSION

The software architecture must support:

Unlimited branches.



Unlimited users.

Additional specialties.

Additional locations.

Multi-country deployment.

EXPECTED DELIVERABLES FROM DEVELOPER

Web Application

Android Application

iOS Application

Admin Dashboard

Database Design

API Documentation

User Manual

Deployment & Hosting Setup



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Staff Training

One-Year Maintenance & Support Plan



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